

DeLeAn Multi-Model Demand-Level Annotation

A Comparative Evaluation Across ADeLe Held-Out Tasks and Novel External Benchmarks

Annotator models: Qwen2.5-72B-Instruct-AWQ · Llama-3.3-70B-Instruct-AWQ · Mistral-Large-Instruct-2411-AWQ (123B)
Distilled model: Qwen2.5-7B, QLoRA fine-tuned on the official ADeLe v1.0 battery

Report generated 2026-07-17

Executive Summary

This report presents two complementary experiments extending the DeLeAn / ADeLe v1.0 demand-level annotation pipeline beyond its original algebra-domain pilot. In both experiments, three large annotator models (Qwen2.5-72B-Instruct-AWQ, Llama-3.3-70B-Instruct-AWQ, and Mistral-Large-Instruct-2411-AWQ) independently score task instances against all 18 ADeLe demand-level rubrics, using identical prompts, decoding settings, and parsing logic, so their outputs are directly comparable.

Part I asks whether the annotation-quality patterns observed on algebra/calculus problems generalize to a genuinely diverse set of task types. We sampled 350 instances (capped at 80 per task) from six ADeLe held-out test tasks that contain no algebra or calculus content whatsoever: MenatQA-Scope, Physics, chemistry, cta (column-type classification), engineering, and zebra_puzzle. The headline result is that the algebra-domain findings do **not** transfer: on algebra, the *minimum* across models was the best de-noising strategy (because all three models shared an upward bias); on this diverse subset, no shared bias exists, and *median/mean* ensembling wins instead. A previously undocumented Mistral failure mode was also identified: on clear-cut factual/procedural tasks, Mistral answers the underlying task question rather than the demand-level rubric question roughly 15-25% of the time, sharply lowering its usable-response rate (74-88% vs. 93-100% for Qwen and Llama).

Part II pushes the test further, to five benchmarks that are entirely absent from the official ADeLe v1.0 battery and were never seen during either the three annotator models' pretraining-scale exposure to ADeLe data or the distilled model's fine-tuning: MBPP (code generation), TheoremQA (theorem-based QA), Omega (geometry/logic puzzles), RelBench (arithmetic/relational reasoning), and SWE-bench-Verified (real GitHub bug patches). This experiment directly targets a concern raised during the project: whether a small (7B) model distilled from the three large annotators' outputs on the official ADeLe training data was simply memorizing surface patterns ("trucado"). The result is unambiguous: the 7B distilled model matches or exceeds all three 70-123B annotator models on this completely unseen benchmark family, winning on mean quadratic-weighted kappa (0.628, tied with Llama-3.3's 0.629 and above Mistral's 0.609 and Qwen's 0.601) and on mean exact-match accuracy (66.6%, clearly ahead of the next-best model at 61.0%). Solo, it also achieves the single best per-dataset exact-match score in all five benchmarks.

Bottom line: (1) ensemble strategy choice must be tuned per task-family, not assumed from a single pilot domain; (2) Mistral's annotation reliability is meaningfully domain-dependent due to an instruction-following failure mode, not just a context-length issue; and (3) the distilled 7B model's strong algebra-domain performance was not an artifact of training-domain overlap — it generalizes cleanly to benchmark families it has never encountered in any form.

Methodology

The ADeLe Framework and the 18 Demand-Level Rubrics

ADeLe (Annotated Demand Levels) v1.0 is an 18-rubric framework for characterizing the cognitive and knowledge demands a task instance places on a solver, independent of whether the solver answers correctly. Each rubric defines six discrete demand levels (0-5) with detailed level descriptions and worked examples. The 18 rubrics span attention/search (AS), atypicality (AT), comprehension (CEc), expression (CEe), conceptualization/ learning/abstraction (CL), four knowledge sub-domains (KNa: applied sciences, KNc: everyday knowledge, KNf: formal sciences, KNn: natural sciences, KNs: social sciences/humanities), three metacognitive rubrics (MCr: identifying relevant information, MCt: critical thinking, MCu: calibrating knowns/unknowns), mind modelling (MS), logical reasoning (QLI), quantitative reasoning (QLq), spatial-physical reasoning (SNs), and volume (VO).

Annotation Procedure

For every (task instance, rubric) pair, each model receives a fixed prompt template: the full rubric text (description + all six level definitions with examples), the task instance itself, and a chain-of-thought instruction asking the model to reason step by step and conclude with the statement "Thus, the level of *X* demanded by the given TASK INSTANCE is: SCORE". All three annotator models were served via vLLM 0.6.1 (offline batch inference, tensor-parallel size 4, AWQ 4-bit quantization, max context length 4096 tokens) on a shared SLURM GPU cluster, using greedy/deterministic decoding so results are reproducible. Model responses are parsed with a regex that extracts the final integer score; responses where no valid score in [0,5] can be extracted are marked unusable and excluded from that model's per-rubric metrics (reported as "% usable").

Evaluation Metrics

- **MAE** — mean absolute error between predicted and gold (reference) level.
- **Exact** — percentage of predictions matching the gold level exactly.
- **Within-1** — percentage of predictions within ± 1 level of gold.
- **κ -quad** — quadratic-weighted Cohen's kappa, the primary ranking metric, since it penalizes larger disagreements more and corrects for chance agreement.

Caveat: κ -quad is uninformative (near 0 or undefined) for rubrics with very low label variance in the gold reference (e.g. KNa, KNc, KNn, KNs, MS often have a large majority class). For these, Exact and MAE are used as the practical tie-breaker instead of trusting κ alone — this logic is applied consistently throughout the "best ensemble candidate" tables below: if the spread of κ -quad across candidates for a given rubric is ≤ 0.02 , ranking falls back to (Exact desc, MAE asc).

Ensemble Strategies

In addition to each model's solo predictions, we evaluate simple, training-free ensembles that combine per-instance predictions across models: **mean**, **median**, **min**, and **max**, computed over various model subsets (pairs and triples, and, in Part II, the full quartet including the distilled model). These require no additional training and test whether simple statistical combination of independent annotators improves reliability, and if so, which combination rule is most robust for a given task family.

Part I — ADeLe Held-Out Task Subset (6 Non-Math Tasks)

Dataset

To test whether patterns found on the algebra/calculus pilot generalize, we sampled 350 instances (capped at 80 per task, using all available instances where a task had fewer) from six ADeLe v1.0 held-out test tasks that contain **no** algebra or calculus content: **MenatQA-Scope**, **Physics**, **chemistry**, **cta** (column-type annotation / classification), **engineering**, and **zebra_puzzle** (constraint-satisfaction logic puzzles). Each instance was scored against all 18 rubrics by all three annotator models ($350 \times 18 = 6,300$ requests per model), against the official ADeLe gold reference level.

Per-Rubric Results (3 Models)

Rubric	Model	n	%Usable	MAE	Exact %	Within-1 %	k-quad
AS (Attention and Search)	Qwen2.5-72B	348	99.4	0.908	36.5	81.6	0.236
AS (Attention and Search)	Llama-3.3-70B	346	98.9	0.74	38.7	88.2	0.444
AS (Attention and Search)	Mistral-Large-12 3B	288	82.3	0.802	38.5	83.7	0.394
AT (Atypicality)	Qwen2.5-72B	344	98.3	0.587	48.5	93.0	0.51
AT (Atypicality)	Llama-3.3-70B	350	100.0	0.577	47.4	94.9	0.57
AT (Atypicality)	Mistral-Large-12 3B	259	74.0	0.71	43.6	88.4	0.32
CEc (Comprehension)	Qwen2.5-72B	346	98.9	0.717	37.3	91.0	0.464
CEc (Comprehension)	Llama-3.3-70B	349	99.7	0.811	37.2	83.4	0.446
CEc (Comprehension)	Mistral-Large-12 3B	300	85.7	0.62	47.3	91.7	0.465
CEe (Expression)	Qwen2.5-72B	349	99.7	0.484	67.6	85.4	0.422
CEe (Expression)	Llama-3.3-70B	350	100.0	0.22	83.7	95.7	0.565
CEe (Expression)	Mistral-Large-12 3B	293	83.7	0.311	75.4	94.9	0.503
CL (Conceptualization, Learning, and Abstraction)	Qwen2.5-72B	344	98.3	0.599	56.4	86.6	0.729
CL (Conceptualization, Learning, and Abstraction)	Llama-3.3-70B	329	94.0	0.912	38.9	76.6	0.519
CL (Conceptualization, Learning, and Abstraction)	Mistral-Large-12 3B	281	80.3	0.982	38.1	75.8	0.441
KNa (Knowledge in Applied Sciences and Professions)	Qwen2.5-72B	348	99.4	0.589	65.5	87.1	0.729
KNa (Knowledge in Applied Sciences and Professions)	Llama-3.3-70B	327	93.4	0.602	59.3	88.4	0.784
KNa (Knowledge in Applied Sciences and Professions)	Mistral-Large-12 3B	291	83.1	0.773	59.1	80.1	0.627
KNc (Customary Everyday Knowledge)	Qwen2.5-72B	327	93.4	0.752	66.1	79.2	0.434
KNc (Customary Everyday Knowledge)	Llama-3.3-70B	349	99.7	0.295	78.5	93.4	0.858
KNc (Customary Everyday Knowledge)	Mistral-Large-12 3B	260	74.3	0.462	79.2	86.5	0.455
KNf (Knowledge in Formal Sciences)	Qwen2.5-72B	348	99.4	0.371	66.4	97.1	0.914
KNf (Knowledge in Formal Sciences)	Llama-3.3-70B	346	98.9	0.509	67.6	89.6	0.82

Rubric	Model	n	%Usable	MAE	Exact %	Within-1 %	k-quad
KNf (Knowledge in Formal Sciences)	Mistral-Large-12 3B	277	79.1	0.718	54.9	84.1	0.676
KNn (Knowledge in Natural Sciences)	Qwen2.5-72B	349	99.7	0.309	78.8	96.0	0.893
KNn (Knowledge in Natural Sciences)	Llama-3.3-70B	339	96.9	0.31	79.1	95.9	0.898
KNn (Knowledge in Natural Sciences)	Mistral-Large-12 3B	291	83.1	0.491	73.9	88.3	0.795
KNs (Knowledge in Social Sciences and Humanities)	Qwen2.5-72B	349	99.7	0.269	83.1	92.6	0.68
KNs (Knowledge in Social Sciences and Humanities)	Llama-3.3-70B	349	99.7	0.206	84.2	96.8	0.776
KNs (Knowledge in Social Sciences and Humanities)	Mistral-Large-12 3B	295	84.3	0.244	86.8	93.2	0.628
MCr (Identifying Relevant Information)	Qwen2.5-72B	331	94.6	0.782	40.8	82.2	0.381
MCr (Identifying Relevant Information)	Llama-3.3-70B	344	98.3	0.593	52.3	89.5	0.547
MCr (Identifying Relevant Information)	Mistral-Large-12 3B	307	87.7	0.893	33.6	80.8	0.31
MCt (Critical Thinking Processes)	Qwen2.5-72B	346	98.9	0.303	69.9	99.7	0.695
MCt (Critical Thinking Processes)	Llama-3.3-70B	333	95.1	0.541	50.5	96.4	0.563
MCt (Critical Thinking Processes)	Mistral-Large-12 3B	288	82.3	0.438	61.1	96.2	0.535
MCu (Calibrating Knowns and Unknowns)	Qwen2.5-72B	343	98.0	0.484	58.3	93.6	0.677
MCu (Calibrating Knowns and Unknowns)	Llama-3.3-70B	337	96.3	0.691	40.1	91.1	0.635
MCu (Calibrating Knowns and Unknowns)	Mistral-Large-12 3B	298	85.1	0.517	58.4	91.3	0.521
MS (Mind Modelling and Social Cognition)	Qwen2.5-72B	350	100.0	0.026	98.9	99.1	-0.002
MS (Mind Modelling and Social Cognition)	Llama-3.3-70B	348	99.4	0.04	96.8	99.1	-0.004
MS (Mind Modelling and Social Cognition)	Mistral-Large-12 3B	281	80.3	0.409	83.6	89.0	-0.002
QLI (Logical Reasoning)	Qwen2.5-72B	346	98.9	0.474	57.5	95.1	0.716
QLI (Logical Reasoning)	Llama-3.3-70B	343	98.0	0.676	49.0	83.7	0.572
QLI (Logical Reasoning)	Mistral-Large-12 3B	290	82.9	0.576	54.5	90.0	0.59
QLq (Quantitative Reasoning)	Qwen2.5-72B	346	98.9	0.289	71.1	100.0	0.917
QLq (Quantitative Reasoning)	Llama-3.3-70B	321	91.7	0.498	57.9	93.8	0.851
QLq (Quantitative Reasoning)	Mistral-Large-12 3B	303	86.6	0.462	63.7	93.4	0.794
SNs (Spatial-physical Reasoning)	Qwen2.5-72B	349	99.7	0.327	77.9	92.3	0.734
SNs (Spatial-physical Reasoning)	Llama-3.3-70B	347	99.1	0.305	80.1	91.6	0.762
SNs (Spatial-physical Reasoning)	Mistral-Large-12 3B	286	81.7	0.524	71.7	84.3	0.522
VO (Volume)	Qwen2.5-72B	349	99.7	0.344	66.5	99.4	0.616

Rubric	Model	n	%Usable	MAE	Exact %	Within-1 %	k-quad
VO (Volume)	Llama-3.3-70B	350	100.0	0.451	56.0	99.1	0.565
VO (Volume)	Mistral-Large-123B	285	81.4	0.512	55.1	95.1	0.488

Table I.1 — Per-rubric annotation quality, 3 annotator models, on the 6-task diverse subset ($n=350$ instances per rubric before usability filtering). Sorted by rubric, then model.

Key Finding: A Mistral-Specific Instruction-Confusion Failure Mode

Mistral's usable-response rate ranges 74.0%-87.7% across rubrics on this subset, markedly lower than Qwen (93.4-100%) and Llama-3.3 (91.7-100%). Manual inspection of the unparseable completions revealed that this is **not** primarily the context-length overflow issue documented on the algebra pilot (where MCr and QLI prompts occasionally exceeded the 4096-token limit). Instead, on task types with a clear "right answer" framing (chemistry QA, cta classification, engineering MCQ), Mistral frequently (roughly 15-25% of responses on affected rubrics) answers the *underlying task question* ("Thus, the correct answer is: X") instead of the *rubric demand-level question* ("Thus, the level of *X* demanded... is: N") that the prompt actually asks for. Qwen and Llama exhibit this confusion far less often (roughly 1-7% of responses).

Ensemble Analysis by Rubric

For each rubric, all pairwise and triple-wise combination strategies were evaluated and the best-performing candidate selected per the tie-break rule above. The table below reports only the winning candidate per rubric; full candidate-level results are in Appendix A.

Rubric	Best Candidate	n	MAE	Exact%	Within-1 %	k-quad
AS (Attention and Search)	median(Q,L,M)	283	0.686	43.1	89.4	0.47
AT (Atypicality)	max(Q,L)	255	0.482	54.9	97.3	0.647
CEc (Comprehension)	min(Q,L,M)	295	0.505	51.9	97.6	0.577
CEe (Expression)	median(Q,L,M)	292	0.195	83.9	96.9	0.7
CL (Conceptualization, Learning, and Abstraction)	qwen (solo)	259	0.602	56.8	85.7	0.72
KNa (Knowledge in Applied Sciences and Professions)	min(Q,L)	277	0.473	71.1	88.8	0.796
KNc (Customary Everyday Knowledge)	llama33 (solo)	238	0.206	85.7	95.8	0.805
KNf (Knowledge in Formal Sciences)	max(Q,L)	272	0.301	71.0	99.3	0.921
KNn (Knowledge in Natural Sciences)	mean(Q,L)	283	0.276	79.2	94.3	0.927
KNs (Knowledge in Social Sciences and Humanities)	median(Q,L,M)	293	0.13	90.8	96.9	0.823
MCr (Identifying Relevant Information)	min(Q,L)	284	0.504	54.9	94.7	0.598
MCT (Critical Thinking Processes)	mean(Q,L,M)	269	0.323	68.0	99.6	0.656
MCu (Calibrating Knowns and Unknowns)	mean(Q,L,M)	279	0.419	61.3	96.8	0.684
MS (Mind Modelling and Social Cognition)	min(Q,L)	280	0.004	99.6	100.0	0.0
QLI (Logical Reasoning)	max(Q,L)	282	0.461	58.5	95.4	0.733
QLq (Quantitative Reasoning)	qwen (solo)	276	0.264	73.6	100.0	0.917
SNs (Spatial-physical Reasoning)	median(Q,L,M)	282	0.27	79.8	94.0	0.816
VO (Volume)	median(Q,L,M)	285	0.298	70.2	100.0	0.689

Table I.2 — Best-performing prediction source per rubric (solo model or ensemble combination), diverse 6-task subset.

Win tally across the 18 rubrics: **median(Q,L,M) wins 5**, max(Q,L) and min(Q,L) win 3 each, qwen-solo and mean(Q,L,M) win 2 each, and min(Q,L,M), llama33-solo, and mean(Q,L) win 1 each. This is qualitatively different from the algebra pilot, where min(Q,L,M)/min(Q,L) dominated (6/17 and 4/17 rubrics respectively) because all three models shared a common upward annotation bias on algebra/calculus problems. On this diverse subset, no single directional bias is shared across models, so *minimum* is no longer specially favored — *median*, which is robust to a single outlier model regardless of direction, becomes the most frequent winner instead.

Per-Task Breakdown

Task	Candidate	n	MAE	Exact%	Within-1 %	k-quad
MenatQA-Scope	qwen (solo)	782	0.459	65.7	91.9	0.669
MenatQA-Scope	llama33 (solo)	782	0.385	67.1	95.9	0.755
MenatQA-Scope	mistral (solo)	782	0.529	62.3	89.4	0.601
MenatQA-Scope	mean(Q,L,M)	782	0.376	67.9	95.5	0.757
MenatQA-Scope	median(Q,L,M)	782	0.349	71.5	95.1	0.763
MenatQA-Scope	min(Q,L,M)	782	0.386	68.4	95.0	0.693
MenatQA-Scope	max(Q,L,M)	782	0.638	55.2	87.1	0.589
Physics	qwen (solo)	1415	0.525	61.9	89.8	0.756
Physics	llama33 (solo)	1415	0.555	55.6	91.1	0.787
Physics	mistral (solo)	1415	0.473	62.3	92.4	0.813
Physics	mean(Q,L,M)	1415	0.426	63.5	94.5	0.843
Physics	median(Q,L,M)	1415	0.417	65.0	94.3	0.844
Physics	min(Q,L,M)	1415	0.53	58.4	90.7	0.78
Physics	max(Q,L,M)	1415	0.606	56.3	88.3	0.737
chemistry	qwen (solo)	1236	0.418	65.7	93.6	0.841
chemistry	llama33 (solo)	1236	0.512	61.7	90.0	0.778
chemistry	mistral (solo)	1236	0.672	58.0	84.5	0.628
chemistry	mean(Q,L,M)	1236	0.436	62.6	94.1	0.834
chemistry	median(Q,L,M)	1236	0.371	69.1	94.8	0.857
chemistry	min(Q,L,M)	1236	0.559	59.9	88.5	0.715
chemistry	max(Q,L,M)	1236	0.672	56.4	84.7	0.681
cta	qwen (solo)	189	0.418	64.6	95.2	0.69
cta	llama33 (solo)	189	0.365	67.2	97.4	0.713
cta	mistral (solo)	189	0.386	68.3	95.2	0.674
cta	mean(Q,L,M)	189	0.27	76.2	96.8	0.792
cta	median(Q,L,M)	189	0.28	75.1	97.4	0.787
cta	min(Q,L,M)	189	0.439	64.6	94.2	0.568
cta	max(Q,L,M)	189	0.45	60.3	96.3	0.709
engineering	qwen (solo)	1241	0.483	63.7	91.5	0.803
engineering	llama33 (solo)	1241	0.567	59.4	87.6	0.78
engineering	mistral (solo)	1241	0.671	56.3	85.0	0.67
engineering	mean(Q,L,M)	1241	0.488	58.0	94.0	0.832
engineering	median(Q,L,M)	1241	0.405	67.4	94.0	0.851
engineering	min(Q,L,M)	1241	0.574	60.9	87.3	0.726

Task	Candidate	n	MAE	Exact%	Within-1 %	k-quad
engineering	max(Q,L,M)	1241	0.743	51.0	82.8	0.686
zebra_puzzle	qwen (solo)	121	0.488	63.6	91.7	0.787
zebra_puzzle	llama33 (solo)	121	0.636	51.2	88.4	0.812
zebra_puzzle	mistral (solo)	121	0.636	57.9	87.6	0.7
zebra_puzzle	mean(Q,L,M)	121	0.471	61.2	91.7	0.843
zebra_puzzle	median(Q,L,M)	121	0.455	67.8	90.9	0.817
zebra_puzzle	min(Q,L,M)	121	0.595	57.0	90.1	0.748
zebra_puzzle	max(Q,L,M)	121	0.711	47.9	86.8	0.745

Table I.3 — Solo models vs. triple ensembles, pooled across all 18 rubrics, broken down by task (n = number of usable (instance, rubric) pairs for that task).

Median(Q,L,M) is the top or near-top candidate in 5 of the 6 tasks (all but zebra_puzzle, where mean(Q,L,M) edges it out marginally). Mistral is the weakest solo model in 5 of 6 tasks, consistent with its elevated unusable-response rate; it is competitive only on Physics.

Discussion

The central conclusion of Part I is that **ensemble strategy choice is task-family dependent and cannot be generalized from a single pilot domain**. The algebra pilot's headline recommendation (use min(Q,L,M) to correct a shared upward bias) does not hold here; a practitioner deploying this pipeline on a new task type should re-run this kind of ensemble sweep rather than assume the algebra-domain best-practice transfers. Separately, Mistral's elevated failure rate on this subset is a genuine, previously undocumented instruction-following weakness (confusing "solve the task" with "annotate the task's demand level") rather than a context-length artifact, and should be accounted for when weighting Mistral's contribution to any production ensemble.

Part II — Novel External Benchmark Evaluation (Including the Distilled Model)

Motivation

A QLoRA-distilled Qwen2.5-7B model was previously fine-tuned to reproduce the demand-level annotation behaviour of the three large annotator models on the official ADeLe v1.0 battery (16,108 items, 63 tasks, 51/6/6 train/val/test task split). Earlier analysis on the algebra domain raised a legitimate concern: the distillation training set included OmniMath Algebra and Calculus items, so strong algebra-domain performance by the distilled model could reflect training-domain overlap rather than genuine demand-level annotation competence. Part II directly addresses this by testing the distilled model, alongside the three large annotators, on five benchmarks that are **completely absent** from the official ADeLe v1.0 battery (verified against the battery's full benchmark list) and were never encountered by the distilled model during fine-tuning in any form.

Dataset

We sampled 300 instances from five external benchmarks with independently-collected ADeLe-style gold annotations: **MBPP** (60 instances, Python code generation), **TheoremQA** (60, theorem-grounded quantitative QA), **Omega** (60, split 30/30 between geometry and logic-puzzle sub-tasks), **RelBench** (60, split 12/12/12/12/12 across addition, locality, science, anagram, and transforms sub-tasks), and **SWE-bench-Verified** (60, real GitHub issue-to-patch software engineering tasks). Each instance was scored against all 18 rubrics by Qwen2.5-72B, Llama-3.3-70B, Mistral-Large-123B, and the distilled Qwen2.5-7B model ($300 \times 18 = 5,400$ requests per model). Rubric prompt templates and rubric content were reconstructed byte-for-byte from the original training data to guarantee identical prompting to all four models.

Per-Rubric Results (4 Models)

Rubric	Model	n	%Usable	MAE	Exact %	Within-1 %	k-quad
AS (Attention and Search)	Qwen2.5-72B	298	99.3	0.822	46.6	80.9	0.436
AS (Attention and Search)	Llama-3.3-70B	299	99.7	0.742	42.8	84.3	0.659
AS (Attention and Search)	Mistral-Large-123B	287	95.7	0.676	44.6	89.5	0.664
AS (Attention and Search)	Distilled-7B	298	99.3	0.55	55.7	89.9	0.735
AT (Atypicality)	Qwen2.5-72B	294	98.0	0.531	53.7	93.5	0.747
AT (Atypicality)	Llama-3.3-70B	297	99.0	0.62	45.5	93.3	0.768
AT (Atypicality)	Mistral-Large-123B	271	90.3	0.701	45.8	89.7	0.578
AT (Atypicality)	Distilled-7B	298	99.3	0.591	48.3	93.6	0.709
CEc (Comprehension)	Qwen2.5-72B	297	99.0	0.67	37.7	95.3	0.653
CEc (Comprehension)	Llama-3.3-70B	297	99.0	0.707	42.1	87.2	0.639
CEc (Comprehension)	Mistral-Large-123B	288	96.0	0.611	45.1	94.4	0.661
CEc (Comprehension)	Distilled-7B	298	99.3	0.393	61.4	99.3	0.767
CEe (Expression)	Qwen2.5-72B	297	99.0	1.125	29.3	66.7	0.229
CEe (Expression)	Llama-3.3-70B	297	99.0	0.987	35.7	76.1	0.253
CEe (Expression)	Mistral-Large-123B	286	95.3	0.79	40.6	83.6	0.34
CEe (Expression)	Distilled-7B	298	99.3	0.691	46.0	85.9	0.368

Rubric	Model	n	%Usable	MAE	Exact %	Within-1 %	k-quad
CL (Conceptualization, Learning, and Abstraction)	Qwen2.5-72B	295	98.3	0.617	51.5	89.8	0.756
CL (Conceptualization, Learning, and Abstraction)	Llama-3.3-70B	296	98.7	0.814	38.2	83.4	0.687
CL (Conceptualization, Learning, and Abstraction)	Mistral-Large-1 23B	274	91.3	0.774	46.7	81.0	0.612
CL (Conceptualization, Learning, and Abstraction)	Distilled-7B	298	99.3	0.594	51.7	89.6	0.776
KNa (Knowledge in Applied Sciences and Professions)	Qwen2.5-72B	296	99.0	1.514	43.6	59.8	0.305
KNa (Knowledge in Applied Sciences and Professions)	Llama-3.3-70B	298	99.7	1.144	56.4	72.8	0.492
KNa (Knowledge in Applied Sciences and Professions)	Mistral-Large-1 23B	284	95.0	1.12	57.0	70.1	0.46
KNa (Knowledge in Applied Sciences and Professions)	Distilled-7B	299	100.0	1.064	65.2	70.6	0.431
KNc (Customary Everyday Knowledge)	Qwen2.5-72B	285	95.0	1.239	52.3	66.7	0.261
KNc (Customary Everyday Knowledge)	Llama-3.3-70B	300	100.0	0.65	69.7	81.7	0.483
KNc (Customary Everyday Knowledge)	Mistral-Large-1 23B	288	96.0	0.729	66.0	78.8	0.357
KNc (Customary Everyday Knowledge)	Distilled-7B	300	100.0	0.56	75.0	82.7	0.503
KNf (Knowledge in Formal Sciences)	Qwen2.5-72B	297	99.0	0.394	62.6	98.3	0.866
KNf (Knowledge in Formal Sciences)	Llama-3.3-70B	299	99.7	0.391	67.9	94.3	0.839
KNf (Knowledge in Formal Sciences)	Mistral-Large-1 23B	292	97.3	0.418	65.8	92.5	0.818
KNf (Knowledge in Formal Sciences)	Distilled-7B	300	100.0	0.333	70.0	96.7	0.869
KNn (Knowledge in Natural Sciences)	Qwen2.5-72B	300	100.0	0.47	82.7	88.3	0.611
KNn (Knowledge in Natural Sciences)	Llama-3.3-70B	300	100.0	0.13	93.7	97.3	0.869
KNn (Knowledge in Natural Sciences)	Mistral-Large-1 23B	292	97.3	0.137	93.8	96.9	0.831
KNn (Knowledge in Natural Sciences)	Distilled-7B	300	100.0	0.173	92.0	96.7	0.783
KNs (Knowledge in Social Sciences and Humanities)	Qwen2.5-72B	299	99.7	0.074	94.6	98.3	0.705
KNs (Knowledge in Social Sciences and Humanities)	Llama-3.3-70B	300	100.0	0.087	95.0	97.7	0.493
KNs (Knowledge in Social Sciences and Humanities)	Mistral-Large-1 23B	295	98.3	0.081	95.6	97.6	0.502
KNs (Knowledge in Social Sciences and Humanities)	Distilled-7B	300	100.0	0.077	97.0	97.3	0.0
MCr (Identifying Relevant Information)	Qwen2.5-72B	293	97.7	0.703	46.8	83.6	0.667
MCr (Identifying Relevant Information)	Llama-3.3-70B	296	98.7	0.905	35.1	78.4	0.612
MCr (Identifying Relevant Information)	Mistral-Large-1 23B	280	93.3	0.786	40.0	84.3	0.601

Rubric	Model	n	%Usable	MAE	Exact %	Within-1 %	k-quad
MCr (Identifying Relevant Information)	Distilled-7B	299	99.7	0.555	51.5	93.6	0.725
MCt (Critical Thinking Processes)	Qwen2.5-72B	297	99.0	0.441	59.6	97.3	0.672
MCt (Critical Thinking Processes)	Llama-3.3-70B	297	99.0	0.485	55.9	95.6	0.695
MCt (Critical Thinking Processes)	Mistral-Large-1 23B	286	95.3	0.399	62.6	97.6	0.707
MCt (Critical Thinking Processes)	Distilled-7B	299	99.7	0.314	68.9	99.7	0.789
MCu (Calibrating Knowns and Unknowns)	Qwen2.5-72B	295	98.3	0.59	50.8	90.5	0.673
MCu (Calibrating Knowns and Unknowns)	Llama-3.3-70B	298	99.3	0.658	42.6	91.9	0.73
MCu (Calibrating Knowns and Unknowns)	Mistral-Large-1 23B	288	96.0	0.618	46.9	91.3	0.633
MCu (Calibrating Knowns and Unknowns)	Distilled-7B	299	99.7	0.545	52.8	93.3	0.721
MS (Mind Modelling and Social Cognition)	Qwen2.5-72B	297	99.0	0.222	92.3	92.6	0.095
MS (Mind Modelling and Social Cognition)	Llama-3.3-70B	299	99.7	0.237	87.0	92.3	0.018
MS (Mind Modelling and Social Cognition)	Mistral-Large-1 23B	279	93.0	0.115	94.3	96.4	0.068
MS (Mind Modelling and Social Cognition)	Distilled-7B	299	99.7	0.043	97.7	98.0	-0.003
QLI (Logical Reasoning)	Qwen2.5-72B	291	97.0	0.498	56.0	94.2	0.761
QLI (Logical Reasoning)	Llama-3.3-70B	297	99.0	0.576	52.2	90.2	0.777
QLI (Logical Reasoning)	Mistral-Large-1 23B	287	95.7	0.519	55.1	94.4	0.733
QLI (Logical Reasoning)	Distilled-7B	298	99.3	0.409	63.8	95.3	0.799
QLq (Quantitative Reasoning)	Qwen2.5-72B	295	98.3	0.454	64.7	92.2	0.788
QLq (Quantitative Reasoning)	Llama-3.3-70B	299	99.7	0.562	56.5	89.0	0.777
QLq (Quantitative Reasoning)	Mistral-Large-1 23B	287	95.7	0.467	62.7	93.0	0.79
QLq (Quantitative Reasoning)	Distilled-7B	299	99.7	0.388	67.6	94.6	0.821
SNs (Spatial-physical Reasoning)	Qwen2.5-72B	297	99.0	0.347	74.4	92.3	0.847
SNs (Spatial-physical Reasoning)	Llama-3.3-70B	297	99.0	0.36	73.7	91.6	0.85
SNs (Spatial-physical Reasoning)	Mistral-Large-1 23B	285	95.0	0.34	73.3	93.0	0.854
SNs (Spatial-physical Reasoning)	Distilled-7B	298	99.3	0.329	75.8	93.3	0.811
VO (Volume)	Qwen2.5-72B	299	99.7	0.385	62.2	99.3	0.745
VO (Volume)	Llama-3.3-70B	300	100.0	0.59	46.3	94.7	0.688
VO (Volume)	Mistral-Large-1 23B	294	98.0	0.398	61.6	98.6	0.75
VO (Volume)	Distilled-7B	299	99.7	0.438	58.5	97.7	0.695

Table II.1 — Per-rubric annotation quality, 4 models (3 large annotators + distilled 7B), on the 5-benchmark novel-task set (n=300 instances per rubric before usability filtering).

Aggregate Summary and the Distillation Question

Model	Mean k-quad	Mean Exact%	Mean Within-1%	Mean %Usable
Llama-3.3-70B	0.629	57.572	88.433	99.4
Distilled-7B	0.628	66.606	92.656	99.628
Mistral-Large-123B	0.609	60.972	90.15	95.25
Qwen2.5-72B	0.601	58.967	87.756	98.572

Table II.2 — Aggregate (mean across 18 rubrics) annotation quality, 5-benchmark novel-task set.

The distilled 7B model achieves the highest mean exact-match accuracy (66.6%) of all four models, " and a mean quadratic-weighted kappa (0.628) statistically tied with the best 70B-scale annotator (Llama-3.3: 0.629), ahead of Mistral-Large-123B (0.609) and Qwen2.5-72B (0.601). Counting per-rubric kappa wins across the 18 rubrics: the distilled model wins outright on **10/18**, Llama-3.3 on 4/18, and Qwen/Mistral on 2/18 each. This is on a benchmark family (code generation, theorem QA, geometry/logic, arithmetic/relational reasoning, and real bug-fix patches) with zero overlap with the distilled model's fine-tuning data — a genuinely out-of-domain test. This result directly counters the concern that the distilled model's strong algebra-domain performance was a training-domain-overlap artifact: on data it has never seen in any form, a model 10-17x smaller than the annotators it was distilled from matches or beats them.

Ensemble Analysis by Rubric

Rubric	Best Candidate	n	MAE	Exact%	Within-1 %	k-quad
AS (Attention and Search)	distilled (solo)	287	0.544	56.4	89.9	0.735
AT (Atypicality)	median(Q,L,M)	268	0.485	56.0	95.5	0.789
CEc (Comprehension)	distilled (solo)	287	0.39	61.3	99.7	0.766
CEe (Expression)	median(Q,L,M,D)	286	0.769	43.4	82.2	0.369
CL (Conceptualization, Learning, and Abstraction)	mean(Q,L,M,D)	273	0.623	46.5	91.6	0.768
KNa (Knowledge in Applied Sciences and Professions)	mean(L,D)	283	1.039	52.7	60.4	0.545
KNc (Customary Everyday Knowledge)	mean(Q,L,M,D)	274	0.77	48.2	81.4	0.502
KNf (Knowledge in Formal Sciences)	median(Q,L,M)	289	0.28	74.0	98.3	0.896
KNn (Knowledge in Natural Sciences)	median(Q,L,M,D)	292	0.113	93.8	96.9	0.892
KNs (Knowledge in Social Sciences and Humanities)	qwen (solo)	295	0.071	94.9	98.3	0.711
MCr (Identifying Relevant Information)	distilled (solo)	278	0.536	52.9	93.9	0.726
MCT (Critical Thinking Processes)	distilled (solo)	283	0.307	69.3	100.0	0.784
MCu (Calibrating Knowns and Unknowns)	min(Q,L,M)	286	0.503	54.2	95.5	0.754
MS (Mind Modelling and Social Cognition)	mean(Q,L,M,D)	279	0.115	90.7	97.8	0.185
QLI (Logical Reasoning)	median(Q,L,M,D)	283	0.399	60.8	99.3	0.834
QLq (Quantitative Reasoning)	median(Q,L,M,D)	283	0.396	66.8	94.0	0.834
SNs (Spatial-physical Reasoning)	median(Q,L,M,D)	285	0.27	77.5	95.4	0.898
VO (Volume)	mean(Q,L,M)	294	0.33	67.3	99.7	0.801

Table II.3 — Best-performing prediction source per rubric (solo model or ensemble, including the distilled model), 5-benchmark novel-task set. Q=Qwen, L=Llama-3.3, M=Mistral, D=Distilled.

Win tally: **median(Q,L,M,D) wins 5/18** and **the distilled model wins solo on 4/18** (i.e. adding it to an ensemble does not always help — on AS, CEc, MCr, and MCt it is already the single best predictor), mean(Q,L,M,D) wins 3/18, median(Q,L,M) (without the distilled model) wins 2/18, and the remainder split 1 win each. Combining all four models continues to help on most rubrics, but the marginal gain over the distilled model alone is far smaller here than the gain ensembling provided on the algebra pilot or the diverse ADeLe subset in Part I — consistent with the distilled model already being a strong, largely-independent predictor rather than a weak one that benefits heavily from correction.

Per-Dataset Breakdown

Dataset	Candidate	n	MAE	Exact%	Within-1 %	k-quad
mbpp	Qwen2.5-72B (solo)	1076	0.588	58.6	88.6	0.637
mbpp	Llama-3.3-70B (solo)	1076	0.497	61.8	91.6	0.703
mbpp	Mistral-Large-123B (solo)	1076	0.516	59.7	92.2	0.707
mbpp	Distilled-7B (solo)	1076	0.439	66.4	93.0	0.73
mbpp	mean(Q,L,M)	1076	0.467	60.5	94.2	0.749
mbpp	median(Q,L,M,D)	1076	0.449	63.1	93.6	0.752
mbpp	mean(Q,L,M,D)	1076	0.455	60.7	94.3	0.765
omega	Qwen2.5-72B (solo)	1046	0.621	59.1	87.9	0.684
omega	Llama-3.3-70B (solo)	1046	0.517	60.2	92.0	0.807
omega	Mistral-Large-123B (solo)	1046	0.572	59.2	88.5	0.742
omega	Distilled-7B (solo)	1046	0.385	68.5	95.5	0.852
omega	mean(Q,L,M)	1046	0.478	60.6	93.0	0.817
omega	median(Q,L,M,D)	1046	0.424	65.1	94.4	0.847
omega	mean(Q,L,M,D)	1046	0.467	59.8	94.4	0.836
relbench	Qwen2.5-72B (solo)	918	0.503	60.1	91.5	0.745
relbench	Llama-3.3-70B (solo)	918	0.496	60.3	91.4	0.743
relbench	Mistral-Large-123B (solo)	918	0.504	61.2	90.7	0.71
relbench	Distilled-7B (solo)	918	0.46	63.1	92.7	0.723
relbench	mean(Q,L,M)	918	0.435	62.3	94.3	0.791
relbench	median(Q,L,M,D)	918	0.42	64.3	94.3	0.79
relbench	mean(Q,L,M,D)	918	0.423	62.9	95.0	0.797
swebenchver	Qwen2.5-72B (solo)	1009	0.724	60.8	83.5	0.627
swebenchver	Llama-3.3-70B (solo)	1009	0.875	47.9	78.6	0.67
swebenchver	Mistral-Large-123B (solo)	1009	0.534	64.9	89.9	0.778
swebenchver	Distilled-7B (solo)	1009	0.475	68.7	90.9	0.799
swebenchver	mean(Q,L,M)	1009	0.635	56.9	87.2	0.749
swebenchver	median(Q,L,M,D)	1009	0.565	62.7	88.4	0.774
swebenchver	mean(Q,L,M,D)	1009	0.597	57.4	88.1	0.786
theoremqa	Qwen2.5-72B (solo)	1056	0.642	57.0	86.8	0.707
theoremqa	Llama-3.3-70B (solo)	1056	0.569	57.9	88.5	0.803
theoremqa	Mistral-Large-123B (solo)	1056	0.545	60.6	90.0	0.782
theoremqa	Distilled-7B (solo)	1056	0.442	68.4	92.3	0.8
theoremqa	mean(Q,L,M)	1056	0.483	60.6	92.6	0.834
theoremqa	median(Q,L,M,D)	1056	0.427	65.2	93.7	0.851
theoremqa	mean(Q,L,M,D)	1056	0.455	60.8	94.7	0.853

Table II.4 — Solo models and ensembles pooled across all 18 rubrics, broken down by external benchmark. n = number of usable (instance, rubric) pairs for that benchmark.

The distilled model achieves the best solo exact-match accuracy in all 5 benchmarks (mbpp: 66.4% vs 58.6-61.8% for the large models; omega: 68.5% vs 59.1-60.2%; relbench: 63.1% vs 60.1-61.2%; swebenchver: 68.7% vs 47.9-64.9%; theoremqa: 68.4% vs 57.0-60.6%), and the best solo k-quad in 4 of 5 (all but relbench, where it is a close second to mean(Q,L,M)). Combining all four models (mean/median) still yields the lowest MAE and highest within-1 agreement per benchmark, but the solo distilled model is already highly competitive — further evidence that its ADeLe-annotation competence is genuine and not domain-overlap-dependent.

Discussion

Part II provides a clean, direct answer to the "is the distilled model rigged?" question raised earlier in this project: tested on five benchmark families with zero presence in its training data, spanning code generation, theorem-grounded mathematics, geometric/logical puzzles, arithmetic/relational reasoning, and real-world software-engineering bug patches, the 7B distilled model is not merely competitive but frequently the single best solo annotator of the four models tested, at a fraction of the parameter count of the 70-123B annotators it was distilled from. This is strong evidence that the distillation captured a genuine, transferable demand-level annotation skill rather than memorized surface statistics from the training tasks. The practical implication is that the distilled model is a credible low-cost substitute for the large annotator ensemble on novel task types, though combining it with even one large model (e.g. median or mean with Llama-3.3) still provides a modest additional accuracy gain in most rubrics.

Overall Conclusions

- **Ensemble strategy must be re-validated per task family.** The algebra-domain recommendation (min(Q,L,M) to correct a shared upward bias) does not hold on diverse non-math tasks, where median/mean-based ensembling wins instead, because the shared directional bias observed on algebra is not present in general.
- **Mistral-Large exhibits a genuine, previously undocumented instruction-following weakness** — confusing the demand-level annotation question with the underlying task question — that is distinct from, and more impactful than, the context-length overflow issue found on the algebra pilot. This elevates its unusable-response rate specifically on tasks with clear "correct answer" framing (chemistry, cta, engineering).
- **The distilled 7B model's competence is genuine and generalizes out-of-domain.** On five external benchmarks with zero overlap with its training data, it matches or beats all three 70-123B annotator models on mean kappa and exact-match, and is the best solo model in the majority of individual rubrics and benchmarks tested. This directly resolves the earlier concern that its algebra-domain results were an artifact of training-domain overlap.
- **Practical recommendation:** for cost-sensitive deployment on novel task types, the distilled model is a credible standalone substitute for the large annotator ensemble; where compute budget allows, combining it with one or two large models via median/mean ensembling still yields a further, if modest, accuracy improvement.

Appendix A — Full Ensemble Candidate Tables

A.1 Part I: All Ensemble Candidates by Rubric (Diverse ADeLe Subset)

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quad
AS	qwen (solo)	283	0.986	34.6	77.7	0.211
AS	llama33 (solo)	283	0.774	36.4	87.3	0.429
AS	mistral (solo)	283	0.806	38.2	83.7	0.399
AS	mean(Q,L)	283	0.735	44.2	83.0	0.352
AS	min(Q,L)	283	0.749	38.2	88.0	0.416
AS	max(Q,L)	283	1.011	32.9	77.0	0.229
AS	mean(Q,L,M)	283	0.717	43.8	85.5	0.396
AS	median(Q,L,M)	283	0.686	43.1	89.4	0.47
AS	min(Q,L,M)	283	0.795	34.6	86.9	0.417
AS	max(Q,L,M)	283	1.085	31.4	72.4	0.191
AT	qwen (solo)	255	0.588	47.8	93.7	0.539
AT	llama33 (solo)	255	0.584	45.9	95.7	0.588
AT	mistral (solo)	255	0.718	43.1	88.2	0.319
AT	mean(Q,L)	255	0.584	46.3	95.3	0.542
AT	min(Q,L)	255	0.69	38.8	92.2	0.486
AT	max(Q,L)	255	0.482	54.9	97.3	0.647
AT	mean(Q,L,M)	255	0.51	50.2	98.8	0.592
AT	median(Q,L,M)	255	0.506	51.4	98.0	0.598
AT	min(Q,L,M)	255	0.792	35.3	86.7	0.395
AT	max(Q,L,M)	255	0.592	50.2	92.9	0.485
CEc	qwen (solo)	295	0.753	34.9	89.8	0.416
CEc	llama33 (solo)	295	0.871	33.9	81.0	0.406
CEc	mistral (solo)	295	0.61	48.1	91.9	0.463
CEc	mean(Q,L)	295	0.827	33.9	83.7	0.41
CEc	min(Q,L)	295	0.627	42.7	94.6	0.522
CEc	max(Q,L)	295	0.997	26.1	76.3	0.337
CEc	mean(Q,L,M)	295	0.627	45.1	92.2	0.495
CEc	median(Q,L,M)	295	0.675	41.4	91.2	0.486
CEc	min(Q,L,M)	295	0.505	51.9	97.6	0.577
CEc	max(Q,L,M)	295	1.054	23.7	73.9	0.303
CEe	qwen (solo)	292	0.534	65.1	82.9	0.415
CEe	llama33 (solo)	292	0.25	81.2	95.2	0.564
CEe	mistral (solo)	292	0.308	75.7	94.9	0.505
CEe	mean(Q,L)	292	0.349	70.2	96.2	0.535
CEe	min(Q,L)	292	0.223	82.5	95.5	0.602
CEe	max(Q,L)	292	0.562	63.7	82.5	0.406
CEe	mean(Q,L,M)	292	0.301	71.9	97.9	0.619
CEe	median(Q,L,M)	292	0.195	83.9	96.9	0.7
CEe	min(Q,L,M)	292	0.257	78.8	95.5	0.491
CEe	max(Q,L,M)	292	0.64	59.2	80.5	0.37

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quad
CL	qwen (solo)	259	0.602	56.8	85.7	0.72
CL	llama33 (solo)	259	1.012	34.7	72.6	0.48
CL	mistral (solo)	259	1.0	37.5	74.9	0.412
CL	mean(Q,L)	259	0.784	37.5	85.3	0.662
CL	min(Q,L)	259	0.931	39.4	74.9	0.517
CL	max(Q,L)	259	0.683	52.1	83.4	0.678
CL	mean(Q,L,M)	259	0.714	41.3	87.6	0.664
CL	median(Q,L,M)	259	0.668	51.4	85.7	0.672
CL	min(Q,L,M)	259	1.077	34.7	68.3	0.419
CL	max(Q,L,M)	259	0.869	42.9	79.2	0.533
KNa	qwen (solo)	277	0.534	67.5	88.8	0.748
KNa	llama33 (solo)	277	0.574	62.1	88.4	0.771
KNa	mistral (solo)	277	0.787	58.8	79.8	0.628
KNa	mean(Q,L)	277	0.516	66.1	86.6	0.796
KNa	min(Q,L)	277	0.473	71.1	88.8	0.796
KNa	max(Q,L)	277	0.635	58.5	88.4	0.722
KNa	mean(Q,L,M)	277	0.57	57.4	89.5	0.793
KNa	median(Q,L,M)	277	0.509	67.1	88.8	0.791
KNa	min(Q,L,M)	277	0.599	69.0	84.5	0.725
KNa	max(Q,L,M)	277	0.787	52.3	83.8	0.618
KNc	qwen (solo)	238	0.567	76.5	84.5	0.399
KNc	llama33 (solo)	238	0.206	85.7	95.8	0.805
KNc	mistral (solo)	238	0.479	78.2	86.1	0.464
KNc	mean(Q,L)	238	0.403	78.2	84.5	0.603
KNc	min(Q,L)	238	0.235	85.3	93.7	0.716
KNc	max(Q,L)	238	0.538	76.9	86.6	0.483
KNc	mean(Q,L,M)	238	0.382	73.5	91.2	0.658
KNc	median(Q,L,M)	238	0.265	85.7	92.4	0.689
KNc	min(Q,L,M)	238	0.248	84.9	93.3	0.635
KNc	max(Q,L,M)	238	0.739	69.7	80.7	0.407
KNf	qwen (solo)	272	0.342	68.4	97.8	0.902
KNf	llama33 (solo)	272	0.548	65.4	89.0	0.761
KNf	mistral (solo)	272	0.71	55.5	84.6	0.681
KNf	mean(Q,L)	272	0.353	72.1	93.0	0.886
KNf	min(Q,L)	272	0.588	62.9	87.5	0.742
KNf	max(Q,L)	272	0.301	71.0	99.3	0.921
KNf	mean(Q,L,M)	272	0.43	64.7	93.0	0.86
KNf	median(Q,L,M)	272	0.335	73.5	95.2	0.879
KNf	min(Q,L,M)	272	0.904	48.9	77.9	0.596
KNf	max(Q,L,M)	272	0.36	66.9	98.2	0.894
KNn	qwen (solo)	283	0.311	77.7	96.5	0.892
KNn	llama33 (solo)	283	0.318	78.4	95.8	0.89
KNn	mistral (solo)	283	0.481	74.2	88.7	0.801
KNn	mean(Q,L)	283	0.276	79.2	94.3	0.927

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quand
KNn	min(Q,L)	283	0.304	79.9	95.8	0.888
KNn	max(Q,L)	283	0.325	76.3	96.5	0.894
KNn	mean(Q,L,M)	283	0.3	76.3	95.1	0.921
KNn	median(Q,L,M)	283	0.237	82.3	96.8	0.927
KNn	min(Q,L,M)	283	0.498	74.9	88.7	0.776
KNn	max(Q,L,M)	283	0.375	73.1	95.4	0.88
KNs	qwen (solo)	293	0.181	89.1	95.2	0.71
KNs	llama33 (solo)	293	0.164	88.1	97.3	0.767
KNs	mistral (solo)	293	0.246	86.7	93.2	0.628
KNs	mean(Q,L)	293	0.177	87.0	95.6	0.776
KNs	min(Q,L)	293	0.133	90.1	97.6	0.772
KNs	max(Q,L)	293	0.212	87.0	94.9	0.719
KNs	mean(Q,L,M)	293	0.181	85.7	96.2	0.786
KNs	median(Q,L,M)	293	0.13	90.8	96.9	0.823
KNs	min(Q,L,M)	293	0.126	90.8	97.6	0.77
KNs	max(Q,L,M)	293	0.334	82.3	91.1	0.586
MCr	qwen (solo)	284	0.81	39.1	81.3	0.365
MCr	llama33 (solo)	284	0.613	51.1	89.1	0.534
MCr	mistral (solo)	284	0.923	32.0	79.6	0.289
MCr	mean(Q,L)	284	0.694	38.0	92.6	0.498
MCr	min(Q,L)	284	0.504	54.9	94.7	0.598
MCr	max(Q,L)	284	0.919	35.2	75.7	0.352
MCr	mean(Q,L,M)	284	0.658	42.3	91.9	0.489
MCr	median(Q,L,M)	284	0.637	46.8	89.4	0.514
MCr	min(Q,L,M)	284	0.511	54.6	94.4	0.558
MCr	max(Q,L,M)	284	1.197	20.8	66.2	0.233
MCt	qwen (solo)	269	0.309	69.5	99.6	0.641
MCt	llama33 (solo)	269	0.513	52.4	97.4	0.541
MCt	mistral (solo)	269	0.439	61.3	95.9	0.526
MCt	mean(Q,L)	269	0.442	56.5	99.3	0.527
MCt	min(Q,L)	269	0.457	55.8	98.9	0.57
MCt	max(Q,L)	269	0.364	66.2	98.1	0.593
MCt	mean(Q,L,M)	269	0.323	68.0	99.6	0.656
MCt	median(Q,L,M)	269	0.327	68.0	99.6	0.642
MCt	min(Q,L,M)	269	0.517	52.0	97.0	0.519
MCt	max(Q,L,M)	269	0.416	63.2	96.3	0.541
MCu	qwen (solo)	279	0.466	60.2	93.5	0.628
MCu	llama33 (solo)	279	0.717	39.4	89.2	0.572
MCu	mistral (solo)	279	0.527	57.7	91.0	0.512
MCu	mean(Q,L)	279	0.613	42.3	96.4	0.62
MCu	min(Q,L)	279	0.649	45.9	89.6	0.57
MCu	max(Q,L)	279	0.534	53.8	93.2	0.624
MCu	mean(Q,L,M)	279	0.419	61.3	96.8	0.684
MCu	median(Q,L,M)	279	0.427	61.6	95.7	0.679

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quad
MCu	min(Q,L,M)	279	0.656	45.9	88.9	0.556
MCu	max(Q,L,M)	279	0.627	49.8	89.2	0.494
MS	qwen (solo)	280	0.032	98.6	98.9	-0.002
MS	llama33 (solo)	280	0.043	96.8	98.9	-0.004
MS	mistral (solo)	280	0.411	83.6	88.9	-0.002
MS	mean(Q,L)	280	0.032	97.5	99.3	-0.004
MS	min(Q,L)	280	0.004	99.6	100.0	0.0
MS	max(Q,L)	280	0.071	95.7	97.9	-0.003
MS	mean(Q,L,M)	280	0.154	87.1	97.5	-0.005
MS	median(Q,L,M)	280	0.032	98.2	98.9	-0.003
MS	min(Q,L,M)	280	0.004	99.6	100.0	0.0
MS	max(Q,L,M)	280	0.45	81.1	87.9	-0.002
QLI	qwen (solo)	282	0.514	54.3	94.3	0.665
QLI	llama33 (solo)	282	0.741	45.4	80.9	0.534
QLI	mistral (solo)	282	0.582	53.9	90.1	0.589
QLI	mean(Q,L)	282	0.56	46.8	97.2	0.66
QLI	min(Q,L)	282	0.794	41.1	79.8	0.459
QLI	max(Q,L)	282	0.461	58.5	95.4	0.733
QLI	mean(Q,L,M)	282	0.5	55.0	95.0	0.68
QLI	median(Q,L,M)	282	0.461	59.2	94.7	0.707
QLI	min(Q,L,M)	282	0.851	38.3	77.3	0.423
QLI	max(Q,L,M)	282	0.525	56.0	93.3	0.663
QLq	qwen (solo)	276	0.264	73.6	100.0	0.917
QLq	llama33 (solo)	276	0.536	55.4	92.8	0.823
QLq	mistral (solo)	276	0.446	64.5	93.8	0.813
QLq	mean(Q,L)	276	0.442	56.9	98.9	0.884
QLq	min(Q,L)	276	0.333	72.8	95.3	0.867
QLq	max(Q,L)	276	0.467	56.2	97.5	0.86
QLq	mean(Q,L,M)	276	0.33	69.9	97.1	0.892
QLq	median(Q,L,M)	276	0.33	71.4	97.1	0.878
QLq	min(Q,L,M)	276	0.373	71.7	93.8	0.834
QLq	max(Q,L,M)	276	0.543	50.4	95.7	0.83
SNs	qwen (solo)	282	0.376	74.8	91.1	0.723
SNs	llama33 (solo)	282	0.365	76.2	90.1	0.748
SNs	mistral (solo)	282	0.525	72.0	84.0	0.512
SNs	mean(Q,L)	282	0.312	79.8	91.1	0.768
SNs	min(Q,L)	282	0.259	81.6	93.6	0.799
SNs	max(Q,L)	282	0.482	69.5	87.6	0.693
SNs	mean(Q,L,M)	282	0.362	69.5	94.7	0.783
SNs	median(Q,L,M)	282	0.27	79.8	94.0	0.816
SNs	min(Q,L,M)	282	0.262	81.2	93.3	0.765
SNs	max(Q,L,M)	282	0.734	62.1	78.0	0.522
VO	qwen (solo)	285	0.351	65.3	99.6	0.624
VO	llama33 (solo)	285	0.46	54.7	99.3	0.577

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-qu ad
VO	mistral (solo)	285	0.512	55.1	95.1	0.488
VO	mean(Q,L)	285	0.34	66.3	99.6	0.607
VO	min(Q,L)	285	0.46	54.7	99.3	0.552
VO	max(Q,L)	285	0.351	65.3	99.6	0.647
VO	mean(Q,L,M)	285	0.302	70.2	99.6	0.678
VO	median(Q,L,M)	285	0.298	70.2	100.0	0.689
VO	min(Q,L,M)	285	0.512	51.6	97.5	0.499
VO	max(Q,L,M)	285	0.512	53.3	96.5	0.507

A.2 Part II: All Ensemble Candidates by Rubric (Novel Benchmarks)

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quad
AS	qwen (solo)	287	0.798	47.4	81.5	0.464
AS	llama33 (solo)	287	0.742	42.9	84.3	0.657
AS	mistral (solo)	287	0.676	44.6	89.5	0.664
AS	distilled (solo)	287	0.544	56.4	89.9	0.735
AS	mean(Q,L,M)	287	0.613	49.8	88.9	0.677
AS	median(Q,L,M)	287	0.575	52.6	90.6	0.708
AS	min(Q,L,M)	287	0.613	49.1	90.9	0.692
AS	max(Q,L,M)	287	1.028	33.1	73.9	0.434
AS	mean(Q,L,M,D)	287	0.585	49.8	91.6	0.716
AS	median(Q,L,M,D)	287	0.585	51.6	90.2	0.702
AS	mean(L,D)	287	0.666	43.9	89.5	0.724
AT	qwen (solo)	268	0.53	54.1	93.3	0.741
AT	llama33 (solo)	268	0.608	46.3	93.7	0.769
AT	mistral (solo)	268	0.698	45.5	89.9	0.588
AT	distilled (solo)	268	0.537	51.1	95.5	0.742
AT	mean(Q,L,M)	268	0.504	54.9	94.8	0.775
AT	median(Q,L,M)	268	0.485	56.0	95.5	0.789
AT	min(Q,L,M)	268	0.515	55.2	95.5	0.75
AT	max(Q,L,M)	268	0.836	34.7	85.8	0.589
AT	mean(Q,L,M,D)	268	0.519	52.2	95.9	0.774
AT	median(Q,L,M,D)	268	0.5	54.1	95.9	0.785
AT	mean(L,D)	268	0.586	46.3	95.5	0.771
CEc	qwen (solo)	287	0.669	37.6	95.5	0.651
CEc	llama33 (solo)	287	0.721	40.8	87.1	0.629
CEc	mistral (solo)	287	0.613	44.9	94.4	0.661
CEc	distilled (solo)	287	0.39	61.3	99.7	0.766
CEc	mean(Q,L,M)	287	0.516	50.5	97.9	0.72
CEc	median(Q,L,M)	287	0.564	46.0	97.6	0.709
CEc	min(Q,L,M)	287	0.561	49.8	94.8	0.676
CEc	max(Q,L,M)	287	0.878	27.5	84.7	0.575
CEc	mean(Q,L,M,D)	287	0.477	53.0	99.3	0.745
CEc	median(Q,L,M,D)	287	0.512	50.5	98.3	0.726
CEc	mean(L,D)	287	0.523	48.8	99.0	0.73
CEe	qwen (solo)	286	1.143	28.7	65.7	0.217
CEe	llama33 (solo)	286	0.986	35.0	75.9	0.259
CEe	mistral (solo)	286	0.79	40.6	83.6	0.34
CEe	distilled (solo)	286	0.689	45.8	86.0	0.369
CEe	mean(Q,L,M)	286	0.895	34.6	79.0	0.309
CEe	median(Q,L,M)	286	0.881	37.8	78.0	0.326
CEe	min(Q,L,M)	286	0.752	42.0	84.3	0.303
CEe	max(Q,L,M)	286	1.287	24.5	62.9	0.206
CEe	mean(Q,L,M,D)	286	0.818	38.1	82.2	0.331

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quad
CEe	median(Q,L,M,D)	286	0.769	43.4	82.2	0.369
CEe	mean(L,D)	286	0.811	42.7	82.9	0.307
CL	qwen (solo)	273	0.63	51.3	89.0	0.739
CL	llama33 (solo)	273	0.842	37.0	82.1	0.662
CL	mistral (solo)	273	0.769	46.9	81.3	0.615
CL	distilled (solo)	273	0.604	50.5	89.7	0.761
CL	mean(Q,L,M)	273	0.637	44.3	92.7	0.756
CL	median(Q,L,M)	273	0.626	49.5	89.4	0.747
CL	min(Q,L,M)	273	0.615	56.0	86.8	0.694
CL	max(Q,L,M)	273	1.0	29.7	76.2	0.593
CL	mean(Q,L,M,D)	273	0.623	46.5	91.6	0.768
CL	median(Q,L,M,D)	273	0.612	49.5	90.5	0.763
CL	mean(L,D)	273	0.747	42.5	84.2	0.714
KNa	qwen (solo)	283	1.551	43.8	58.3	0.294
KNa	llama33 (solo)	283	1.159	56.9	72.4	0.488
KNa	mistral (solo)	283	1.124	56.9	70.0	0.459
KNa	distilled (solo)	283	1.057	66.8	70.7	0.418
KNa	mean(Q,L,M)	283	1.226	38.5	69.6	0.476
KNa	median(Q,L,M)	283	1.17	56.5	70.0	0.458
KNa	min(Q,L,M)	283	0.961	64.7	73.5	0.516
KNa	max(Q,L,M)	283	1.703	36.4	57.2	0.296
KNa	mean(Q,L,M,D)	283	1.113	38.2	67.8	0.543
KNa	median(Q,L,M,D)	283	1.046	55.1	68.2	0.526
KNa	mean(L,D)	283	1.039	52.7	60.4	0.545
KNc	qwen (solo)	274	1.248	52.9	66.1	0.261
KNc	llama33 (solo)	274	0.646	70.4	81.8	0.481
KNc	mistral (solo)	274	0.73	65.7	78.8	0.371
KNc	distilled (solo)	274	0.588	74.5	81.8	0.472
KNc	mean(Q,L,M)	274	0.803	46.7	79.9	0.499
KNc	median(Q,L,M)	274	0.675	69.0	80.7	0.465
KNc	min(Q,L,M)	274	0.646	70.8	81.0	0.363
KNc	max(Q,L,M)	274	1.303	49.3	65.0	0.282
KNc	mean(Q,L,M,D)	274	0.77	48.2	81.4	0.502
KNc	median(Q,L,M,D)	274	0.646	67.9	79.9	0.483
KNc	mean(L,D)	274	0.599	73.0	80.7	0.502
KNf	qwen (solo)	289	0.398	62.3	98.3	0.859
KNf	llama33 (solo)	289	0.384	68.5	94.5	0.835
KNf	mistral (solo)	289	0.419	65.7	92.4	0.817
KNf	distilled (solo)	289	0.336	69.6	96.9	0.863
KNf	mean(Q,L,M)	289	0.308	71.6	97.9	0.884
KNf	median(Q,L,M)	289	0.28	74.0	98.3	0.896
KNf	min(Q,L,M)	289	0.585	52.9	90.0	0.755
KNf	max(Q,L,M)	289	0.336	69.6	96.9	0.871
KNf	mean(Q,L,M,D)	289	0.336	68.5	97.9	0.879

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quad
KNf	median(Q,L,M,D)	289	0.318	69.9	98.3	0.887
KNf	mean(L,D)	289	0.37	66.4	96.9	0.862
KNn	qwen (solo)	292	0.476	82.9	88.0	0.593
KNn	llama33 (solo)	292	0.134	93.5	97.3	0.858
KNn	mistral (solo)	292	0.137	93.8	96.9	0.831
KNn	distilled (solo)	292	0.161	93.2	96.9	0.771
KNn	mean(Q,L,M)	292	0.233	83.6	94.9	0.847
KNn	median(Q,L,M)	292	0.13	93.8	97.3	0.858
KNn	min(Q,L,M)	292	0.113	95.2	97.6	0.842
KNn	max(Q,L,M)	292	0.503	81.2	87.3	0.597
KNn	mean(Q,L,M,D)	292	0.212	83.2	96.6	0.871
KNn	median(Q,L,M,D)	292	0.113	93.8	96.9	0.892
KNn	mean(L,D)	292	0.14	92.8	96.2	0.851
KNs	qwen (solo)	295	0.071	94.9	98.3	0.711
KNs	llama33 (solo)	295	0.088	94.9	97.6	0.493
KNs	mistral (solo)	295	0.081	95.6	97.6	0.502
KNs	distilled (solo)	295	0.078	96.9	97.3	0.0
KNs	mean(Q,L,M)	295	0.078	93.6	99.0	0.647
KNs	median(Q,L,M)	295	0.068	95.6	98.6	0.532
KNs	min(Q,L,M)	295	0.068	96.9	97.6	0.29
KNs	max(Q,L,M)	295	0.105	92.9	97.3	0.703
KNs	mean(Q,L,M,D)	295	0.061	95.9	98.3	0.61
KNs	median(Q,L,M,D)	295	0.068	96.6	98.0	0.35
KNs	mean(L,D)	295	0.078	95.9	97.3	0.335
MCr	qwen (solo)	278	0.716	46.4	82.7	0.652
MCr	llama33 (solo)	278	0.906	34.9	78.4	0.61
MCr	mistral (solo)	278	0.788	39.9	84.2	0.601
MCr	distilled (solo)	278	0.536	52.9	93.9	0.726
MCr	mean(Q,L,M)	278	0.712	43.2	86.0	0.664
MCr	median(Q,L,M)	278	0.701	45.3	85.6	0.662
MCr	min(Q,L,M)	278	0.676	46.4	87.4	0.644
MCr	max(Q,L,M)	278	1.032	29.5	72.3	0.574
MCr	mean(Q,L,M,D)	278	0.626	45.0	92.8	0.725
MCr	median(Q,L,M,D)	278	0.64	45.7	90.6	0.707
MCr	mean(L,D)	278	0.763	34.9	91.0	0.691
MCt	qwen (solo)	283	0.428	59.4	97.9	0.704
MCt	llama33 (solo)	283	0.473	56.5	96.1	0.691
MCt	mistral (solo)	283	0.403	62.2	97.5	0.692
MCt	distilled (solo)	283	0.307	69.3	100.0	0.784
MCt	mean(Q,L,M)	283	0.364	64.7	98.9	0.731
MCt	median(Q,L,M)	283	0.353	65.7	98.9	0.735
MCt	min(Q,L,M)	283	0.375	64.7	97.9	0.74
MCt	max(Q,L,M)	283	0.576	47.7	94.7	0.627
MCt	mean(Q,L,M,D)	283	0.392	61.5	99.3	0.724

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quad
MCt	median(Q,L,M,D)	283	0.399	60.8	99.3	0.72
MCt	mean(L,D)	283	0.452	56.2	98.6	0.711
MCu	qwen (solo)	286	0.598	50.3	90.2	0.661
MCu	llama33 (solo)	286	0.661	42.7	91.6	0.721
MCu	mistral (solo)	286	0.622	46.5	91.3	0.63
MCu	distilled (solo)	286	0.538	52.8	93.4	0.728
MCu	mean(Q,L,M)	286	0.514	52.8	95.8	0.734
MCu	median(Q,L,M)	286	0.531	52.1	94.8	0.715
MCu	min(Q,L,M)	286	0.503	54.2	95.5	0.754
MCu	max(Q,L,M)	286	0.846	33.2	82.9	0.582
MCu	mean(Q,L,M,D)	286	0.549	48.3	96.9	0.737
MCu	median(Q,L,M,D)	286	0.552	48.3	96.5	0.735
MCu	mean(L,D)	286	0.661	38.8	95.1	0.732
MS	qwen (solo)	279	0.229	92.1	92.5	0.098
MS	llama33 (solo)	279	0.222	87.1	92.8	-0.03
MS	mistral (solo)	279	0.115	94.3	96.4	0.068
MS	distilled (solo)	279	0.036	98.2	98.2	0.0
MS	mean(Q,L,M)	279	0.172	86.4	96.4	0.157
MS	median(Q,L,M)	279	0.104	93.9	96.4	0.1
MS	min(Q,L,M)	279	0.036	98.2	98.2	0.0
MS	max(Q,L,M)	279	0.427	81.4	87.1	0.075
MS	mean(Q,L,M,D)	279	0.115	90.7	97.8	0.185
MS	median(Q,L,M,D)	279	0.061	96.1	97.8	0.104
MS	mean(L,D)	279	0.104	92.8	96.8	-0.03
QLI	qwen (solo)	283	0.502	55.8	94.0	0.751
QLI	llama33 (solo)	283	0.58	51.9	90.1	0.771
QLI	mistral (solo)	283	0.516	55.5	94.3	0.733
QLI	distilled (solo)	283	0.399	64.7	95.4	0.798
QLI	mean(Q,L,M)	283	0.41	60.8	98.2	0.825
QLI	median(Q,L,M)	283	0.392	63.3	97.5	0.824
QLI	min(Q,L,M)	283	0.565	55.8	89.0	0.683
QLI	max(Q,L,M)	283	0.64	44.2	91.9	0.757
QLI	mean(Q,L,M,D)	283	0.438	56.5	99.6	0.822
QLI	median(Q,L,M,D)	283	0.399	60.8	99.3	0.834
QLI	mean(L,D)	283	0.53	48.8	98.2	0.803
QLq	qwen (solo)	283	0.459	64.3	92.2	0.777
QLq	llama33 (solo)	283	0.555	56.9	89.4	0.771
QLq	mistral (solo)	283	0.463	62.9	93.3	0.791
QLq	distilled (solo)	283	0.378	67.8	95.4	0.822
QLq	mean(Q,L,M)	283	0.449	60.4	94.7	0.827
QLq	median(Q,L,M)	283	0.431	64.0	94.0	0.821
QLq	min(Q,L,M)	283	0.396	70.0	92.9	0.803
QLq	max(Q,L,M)	283	0.65	50.2	88.0	0.721
QLq	mean(Q,L,M,D)	283	0.413	64.3	94.3	0.833

Rubric	Candidate	n	MAE	Exact %	Within-1 %	k-quad
QLq	median(Q,L,M,D)	283	0.396	66.8	94.0	0.834
QLq	mean(L,D)	283	0.438	62.5	94.0	0.819
SNs	qwen (solo)	285	0.34	74.0	92.6	0.862
SNs	llama33 (solo)	285	0.368	73.3	91.2	0.849
SNs	mistral (solo)	285	0.34	73.3	93.0	0.854
SNs	distilled (solo)	285	0.326	75.1	93.3	0.833
SNs	mean(Q,L,M)	285	0.284	75.8	95.8	0.895
SNs	median(Q,L,M)	285	0.281	78.2	94.0	0.885
SNs	min(Q,L,M)	285	0.281	77.9	94.4	0.872
SNs	max(Q,L,M)	285	0.488	64.6	88.4	0.816
SNs	mean(Q,L,M,D)	285	0.274	76.8	95.8	0.895
SNs	median(Q,L,M,D)	285	0.27	77.5	95.4	0.898
SNs	mean(L,D)	285	0.291	75.8	95.1	0.884
VO	qwen (solo)	294	0.381	62.6	99.3	0.743
VO	llama33 (solo)	294	0.599	45.6	94.6	0.677
VO	mistral (solo)	294	0.398	61.6	98.6	0.75
VO	distilled (solo)	294	0.439	58.5	97.6	0.688
VO	mean(Q,L,M)	294	0.33	67.3	99.7	0.801
VO	median(Q,L,M)	294	0.34	66.3	99.7	0.795
VO	min(Q,L,M)	294	0.531	52.0	94.9	0.66
VO	max(Q,L,M)	294	0.507	51.4	98.0	0.708
VO	mean(Q,L,M,D)	294	0.357	65.0	99.3	0.774
VO	median(Q,L,M,D)	294	0.378	62.9	99.3	0.763
VO	mean(L,D)	294	0.51	51.4	97.6	0.71